

Experiment 0

Aim: To study SQL commands, their uses, syntax, and install MySQL for database management experiments.

What is SQL?

SQL is a standard language for accessing and manipulating databases.

- SQL stands for **Structured Query Language**
- SQL lets you **access and manipulate** databases
- SQL became a standard of the American National Standards Institute (**ANSI**) in 1986, and of the International Organization for Standardization (**ISO**) in 1987

Uses of SQL

- Create and manage databases.
- Insert, update, and delete records.
- Retrieve data using queries.
- Control user access and permissions.
- Maintain data integrity and relationships between tables.

SQL Commands: SQL commands are classified into five categories:

DDL (Data Definition Language) - Used to define and modify database objects.

Command	Use	Syntax
CREATE	Creates a table or database	CREATE TABLE table_name (...);
ALTER	Modifies an existing table	ALTER TABLE table_name ADD column_name datatype;
DROP	Deletes a table or database	DROP TABLE table_name;
TRUNCATE	Removes all rows from a table	TRUNCATE TABLE table_name;

1. DML (Data Manipulation Language)

Used to manipulate data stored in tables.

Command	Use	Syntax
INSERT	Adds new records	INSERT INTO table_name VALUES (...);
UPDATE	Modifies existing records	UPDATE table_name SET column=value WHERE condition;
DELETE	Removes records	DELETE FROM table_name WHERE condition;

2. DQL (Data Query Language)

Command	Use	Syntax
SELECT	Retrieves data from a table	SELECT * FROM table_name;

Common clauses used with SELECT:

- WHERE – Filters records.
- ORDER BY – Sorts records.
- GROUP BY – Groups similar records.
- HAVING – Filters grouped results.
- DISTINCT – Removes duplicate values.

3. DCL (Data Control Language)

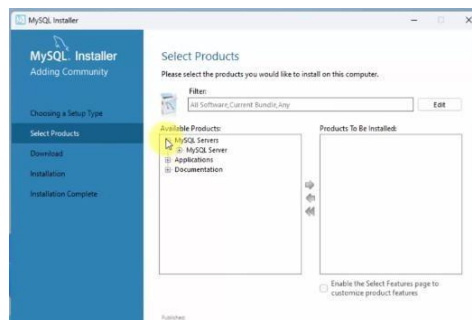
Command	Use	Syntax
GRANT	Gives user privileges	GRANT privilege ON table TO user;
REVOKE	Removes privileges	REVOKE privilege ON table FROM user;

MySQL Installation Procedure

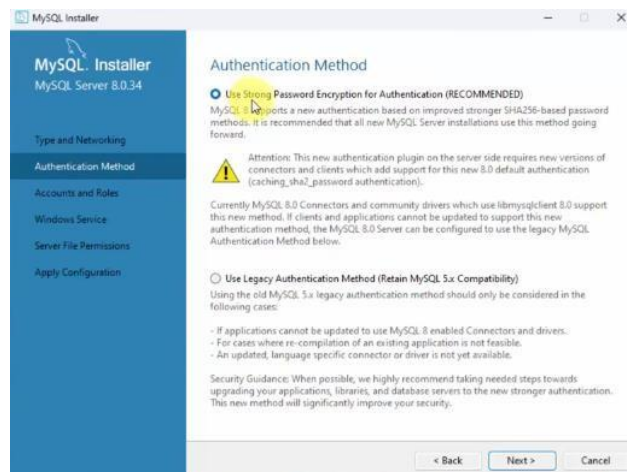
- Download the **MySQL Installer Community Edition**.



- Run Installer: Open the downloaded MSI file to begin the setup.

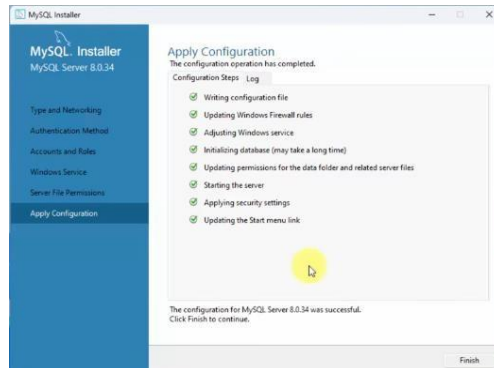


- Setup Type: Choose the Custom setup type to have control over which products are installed.

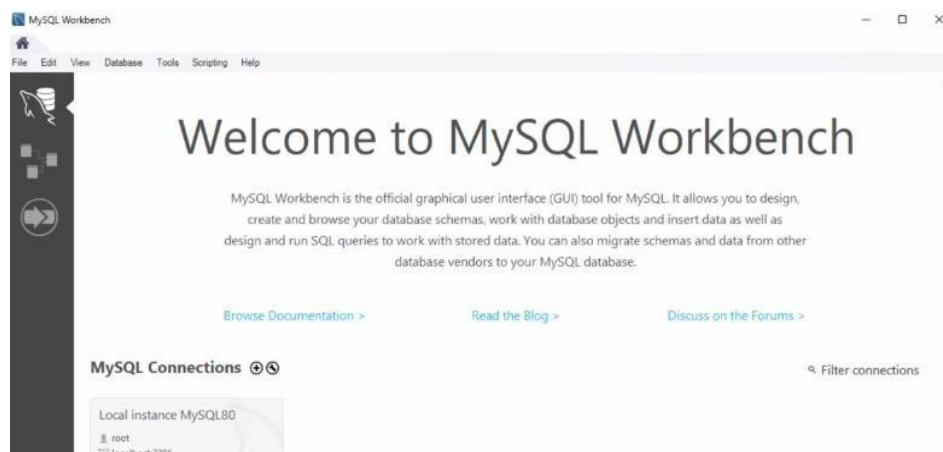


- Create a password for the **root** user.
- Keep the default Windows service settings.

- Apply the configuration and complete the installation.



- Launch **MySQL Workbench**.



- Connect using:
 - **Host:** localhost
 - **Username:** root
 - **Password:** (created during installation)
- Verify the installation by executing:

SHOW DATABASES;

If the list of databases is displayed, MySQL has been installed successfully.

Conclusion: SQL is an essential language for working with relational databases. It provides simple commands to create, manage, retrieve, and manipulate data efficiently. Installing MySQL provides a practical environment to execute SQL queries and develop database applications.